



Department of Computer Science
COMP2421 (Spring 2025/2026)
Project#2 Due date: 21 June 2025 (before 20:00)

In this project, you will build a system to maintain the information of buildings and their information. Your program should read a set of building names and their relevant information from a file named info.txt

The format of the input data is as follows:

Name:number:address:no. of appartments:establishemt
year:Payed fees

Example

Alquds:20:Al Marj:15:1999:yes
Jbenah:121:Nishan street:15:2012:no
Iskan Al Andalus:120:Gabi Baramke:10:2020:no
Al Rabee 1:512:Rabee Main Street:10:2021:yes
Al Rabee 2:513:Rabee Main Street:25:2024:yes

You are required to implement a program to help maintaining the information of buildings using an AVL Tree:

- Read info.txt file and create the tree.
- Insert a new building from the user with all its associated information.
- Find a building and give the user the option to update its information (if found). Update the building number, address, number of apartments, and if they paid all fees.
- List the buildings in the tree in alphabetical order with their associated information.
- List all buildings that have a number of apartments greater than an input number.
- List all buildings that have not paid their fees.
- Delete a building from the tree.

- Save all information to a file called `hash.txt`.

Then create a Hash Table using the data from `hash.txt`. Use the name of buildings as the keys and implement the following functions on the hash table:

- Print hashed table (i.e., print the entire table to the screen including empty spots).
- Print the size of hash table and the load factor.
- Insert a new record into the hash table.
- Search for a specific building and print the number of collisions to reach the building (if there are any collisions).
- Delete a specific record.
- Exit and Save the data back to the file `hash.txt`.

Please note that you should use any open addressing hashing technique and you should use only the first 4 characters of the name of building as the key.

The deadline of this project is on Sunday 21 June 2026 before 8:00PM. Any late submission after 8:00PM to the midnight will be penalized 30%, till 12:00PM the next day will be penalized 70%, and after will not be accepted. Please make sure that your application is running properly on your laptop before the discussion. Project discussions will be determined by your instructor.

Grading policy and general notes on the project:

1. Your application should have all functionalities working properly. Twenty marks will be graded for the functionality of the project.
2. The following notes will make up the remaining 10 marks of the grade:
 - a. There has to be adequate documentation and comments in the code (i.e., functions, loops, etc.). Your code should follow the code convention (i.e., spaces, indentations, etc.); and
 - b. Your application should contain a menu to allow the user to select which option(s).

Notes and submission instructions:

1. **This is individual work.** It should represent your own efforts. It is fine to discuss your work and to ask your colleagues, but you are not allowed to copy/paste the work of others or give your work to anyone else. You are not allowed to post/copy from other websites and/or social media and this will be considered as cheating.
2. Any **plagiarized** code will not be marked.
3. **Document format.** Please submit only the code file (**c** file) containing the code of your project. Please rename it as follows:
“**P2_YourStudentID_FirstNameLastName_SectionNo.c**”.
4. **Input/output file name.** Make sure that the input/output file names are the same as in the specifications.
5. Include your full name, student ID, and section number in the beginning of your file.
6. Please do not compress the file, only the C-file is needed.
7. Files not following the naming convention in point 2 will be penalized.